

CSC-324: Computer Organization and Assembly Language (LAB)

General Information

Course Number	CSC-324
Credit Hours	3(2+1) (Theory Credit Hour = 3, Lab Credit Hours = 1)
Prerequisite	Digital Logic Design
Course Instructor	Bilal Ahmed

Course Objectives

This course introduces the organization of digital computer systems, focusing on how hardware components and assembly-level programming interact. Major topics include the organization of processor, memory, and input/output modules, as well as the interconnections among these components. The course examines the central processing unit and its functional units, including the role of registers and their organization. Students will gain an understanding of instruction formats and the basic instruction sets of 16-bit and 32-bit processors. Emphasis is placed on developing skills in assembly language programming, including program structure, instruction usage, and programming techniques, to bridge theoretical organization concepts with practical implementation. Understand the functions of Registers in different dimension of computer organization in continuation with organization and architecture, Students should learn assembly language programming structure & techniques using variables, statements, mnemonics, control structures using jumps and branches, arrays and data structures, together with the basic structured programming techniques of sequence, choice and iteration and how they can be realized in assembly language. Students should examine the logic bits operations.

Catalog Description

CSC 324

Week No.	Lab Title	Activities / Subtopics	CLO
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1-2	Introduction to Assembly Language	Program structure, data & variable definitions, memory models, MOV & XCHG instructions	1
2-3	Arithmetic Instructions	ADD, SUB, INC, DEC, NEG, writing small arithmetic programs	1
3-5	Input / Output Programming	I/O using INT 21h, INT 10h, INT 16h Writing first program (create & run output)	2
5-7	Flow Control	Conditional & unconditional jumps, branching and looping programs	2
8	Logical Instructions	AND, OR, XOR, NOT, TEST instructions with examples	2
9	Shift and Rotate Instructions	Logical/arithmetic shifts, rotate left/right, simple applications	2
9-11	Stack Operations	PUSH & POP instructions, CALL & RET, stack-based procedure writing	3
12	Arrays	One- and two-dimensional arrays, element access and manipulation	3
13-14	String Operations	MOVS, STOS, LODS, SCAS, CMPS instructions and applications	3
15--	Mini Project / Review	Integration of learned concepts in a small assembly project	–

Text Book

1. Computer Organization and Architecture By: William Stallings Prentice Hall
2. Ytha YU and Charles Marut, “Assembly Language Programming and Organization of the IBM PC”

Reference Material

1. Kip R Irvine, “Assembly Language for Intel-based Computers”
2. Essentials of Computer Architecture By: Douglas E. Comer

Course Learning Outcomes

Course Learning Outcomes (CLO)	
1	Understanding the nature and characteristics of modern computer systems architecture with employed Low level programming tools and techniques.
2	To Apply low-level programming logic employed for problem solving with respect to organization of computer system.

